MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 08 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The sum of the first n positive natural numbers is: [1]

(A) $\frac{n(n-1)}{2}$ (B) $\frac{n(n+1)}{2}$ (C) $n(n+1)$ (D) $\frac{n^2}{2}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $\frac{n(n+1)}{2}$** [0.5 Mark]

Given: First n positive natural numbers $1, 2, 3, \ldots, n$ ($a = 1$, $d = 1$, $l = n$).

Formula / Theorem Used: $S_n = \frac{n}{2}(a + l)$

Justification: [0.5 Mark]

$S_n = \frac{n}{2}(1 + n) = \frac{\mathbf{n(n+1)}}{\mathbf{2}}$.

Q2. The sum of the first n odd positive natural numbers is: [1]

(A) $2n - 1$ (B) n^2 (C) $2n^2$ (D) $n(n + 1)$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) $\mathbf{n^2}$** [0.5 Mark]

Given: Odd natural numbers $1, 3, 5, \ldots$ form an AP with $a = 1$, $d = 2$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

$$S_n = \frac{n}{2}[2(1) + (n-1)(2)] = \frac{n}{2}(2 + 2n - 2) = \frac{n}{2}(2n) = \mathbf{n^2}$$

Q3. The sum of the first n positive even integers is: [1]

(A) n^2 (B) $n(n+1)$ (C) $n(n-1)$ (D) $\dfrac{n(n+1)}{2}$

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) $\mathbf{n(n+1)}$** [0.5 Mark]

Given: Even positive integers $2, 4, 6, \ldots$ form an AP with $a = 2$, $d = 2$.

Formula / Theorem Used: $S_n = \dfrac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

$$S_n = \frac{n}{2}[2(2) + (n-1)(2)] = \frac{n}{2}(4 + 2n - 2) = \frac{n}{2}(2n+2) = \mathbf{n(n+1)}$$

Q4. If the 9th term of an AP is zero ($a_9 = 0$), then the ratio $a_{29} : a_{19}$ is: [1]

(A) 1 : 2 (B) 2 : 1 (C) 3 : 1 (D) 1 : 3

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(B) 2 : 1** [0.5 Mark]

We apply the general term formula $a_n = a + (n-1)d$:

Formula / Theorem Used: $a_9 = a + 8d = 0 \implies a = -8d$

Justification: [0.5 Mark]

$a_{29} = a + 28d = -8d + 28d = 20d$ and $a_{19} = a + 18d = -8d + 18d = 10d$.

Hence $\frac{a_{29}}{a_{19}} = \frac{20d}{10d} = \frac{2}{1} \implies a_{29} : a_{19} = \mathbf{2 : 1}$.

Q5. **Assertion (A):** If the nth term of a sequence is $a_n = 2n^2 + 1$, the sequence forms an AP.
Reason (R): A sequence is an AP if and only if its nth term is a linear expression in n. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(D) Assertion (A) is false, but Reason (R) is true.** [0.5 Mark]

We check linearity of a_n and test consecutive differences:

Formula / Theorem Used: $a_n = a + (n-1)d = dn + (a-d)$ (always linear in n)

Justification: [0.5 Mark]

For $a_n = 2n^2 + 1$: $a_1 = 3$, $a_2 = 9$, $a_3 = 19$. Here $a_2 - a_1 = 6 \neq 10 = a_3 - a_2$, so it is *not* an AP (Assertion (A) is false). Reason (R) states the true necessary and sufficient condition for an AP. Hence **(D)**.

Q6. Find the sum of all natural numbers between 100 and 500 that are divisible by 8. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying the AP and Finding n [1 Mark]

First multiple of 8 strictly greater than 100 is 104 ($8 \times 13 = 104$).
Last multiple of 8 strictly less than 500 is 496 ($8 \times 62 = 496$).
The AP is $104, 112, 120, \ldots, 496$ with $a = 104$, $d = 8$, and $a_n = 496$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

$$104 + (n-1)(8) = 496 \implies 8(n-1) = 496 - 104 = 392 \implies n - 1 = \frac{392}{8} = 49 \implies n = 50$$

Step 2: Computing the Sum S_{50} [1 Mark]

$$S_{50} = \frac{50}{2}(104 + 496) = 25 \times 600 = \boxed{\mathbf{15{,}000}}$$

Q7. If S_1, S_2, S_3 are the sums of n, $2n$, and $3n$ terms respectively of an AP, prove that $S_3 = 3(S_2 - S_1)$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: First term $= a$, common difference $= d$. Let S_k be the sum of k terms:

Formula / Theorem Used: $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Writing Expressions for S_1, S_2, and S_3 [1 Mark]

$$S_1 = \frac{n}{2}[2a + (n-1)d], \quad S_2 = \frac{2n}{2}[2a + (2n-1)d], \quad S_3 = \frac{3n}{2}[2a + (3n-1)d]$$

Step 2: Simplifying RHS $= 3(S_2 - S_1)$ [2 Marks]

$$\begin{aligned} 3(S_2 - S_1) &= 3 \cdot \frac{n}{2}\big[2(2a + (2n-1)d) - (2a + (n-1)d)\big] \\ &= \frac{3n}{2}[4a + (4n-2)d - 2a - (n-1)d] \\ &= \frac{3n}{2}[2a + (4n - 2 - n + 1)d] = \frac{3n}{2}[2a + (3n-1)d] = \boxed{\mathbf{S_3} \quad \text{[Hence Proved]}} \end{aligned}$$

Q8. Find four numbers in AP whose sum is 20 and the sum of whose squares is 120. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Choose four consecutive AP terms symmetrically as:

Formula / Theorem Used: Four numbers in AP: $(a-3d)$, $(a-d)$, $(a+d)$, $(a+3d)$

Step 1: Finding Central Value a [1 Mark]

$(a-3d) + (a-d) + (a+d) + (a+3d) = 20 \implies 4a = 20 \implies a = 5$.

Step 2: Applying Sum of Squares $= 120$ [1.5 Marks]

$$(5-3d)^2 + (5-d)^2 + (5+d)^2 + (5+3d)^2 = 120$$

Using $(A-B)^2 + (A+B)^2 = 2(A^2 + B^2)$:

$$2(25+9d^2)+2(25+d^2) = 120 \implies 50+18d^2+50+2d^2 = 120 \implies 20d^2 = 20 \implies d^2 = 1 \implies d = \pm 1$$

Step 3: Listing the Four Numbers [0.5 Mark]

For $d = 1$: $2, 4, 6, 8$; for $d = -1$: $8, 6, 4, 2$.

Four numbers: **2**, **4**, **6**, **8** (or **8**, **6**, **4**, **2**)

Q9. In an AP, the first term is 5, the last term is 45, and the sum of all terms is 400. Find the number of terms and the common difference. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a = 5$, $l = a_n = 45$, and $S_n = 400$.

Formula / Theorem Used: $S_n = \frac{n}{2}(a + l)$ and $l = a + (n-1)d$

Step 1: Finding the Number of Terms (n) [1.5 Marks]

$$400 = \frac{n}{2}(5 + 45) \implies 400 = \frac{50n}{2} = 25n \implies n = \frac{400}{25} = 16$$

Step 2: Finding the Common Difference (d) [1.5 Marks]

Substituting $n = 16$, $a = 5$, $l = 45$ into $l = a + (n-1)d$:

$$45 = 5 + (16 - 1)d \implies 15d = 45 - 5 = 40 \implies d = \frac{40}{15} = \frac{8}{3}$$

$$\boxed{\mathbf{n = 16}, \qquad \mathbf{d = \frac{8}{3}}}$$

MATHS MANTHAN TUITION CLASSES OFFICIAL SOLUTIONS

Q10. CASE STUDY: Tiered Grandstand Seating in a Sports Stadium

A newly built municipal cricket stadium features a semicircular spectator stand constructed in 30 ascending tiers. The 1st (bottommost) row holds 200 seats, the 2nd row holds 220 seats, the 3rd row holds 240 seats, and each successive row increases by 20 seats right up to the 30th tier: [4]

(i) Write the AP representing row-wise seating capacities and state its a and d. [1]

(ii) Calculate the number of seats available in the 15th row. [1]

(iii) Find the total seating capacity of the entire 30-tier grandstand. [2]

— **OR** —

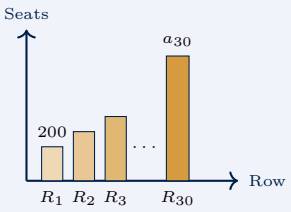
If tickets for rows 1 to 10 are priced at Rs. 500 each, compute the total box-office revenue from those first 10 rows when full. [2]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Row capacities form an AP starting at $a = 200$ with uniform increment $d = 20$ over $n = 30$ tiers.
We apply the AP term and sum formulas:

Formula / Theorem

Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$



Sub-part (i): Seating Capacity AP, a, and d [1 Mark]

The sequence of seats is **200, 220, 240, ..., 780**, where **a = 200** and **d = 20**.

Sub-part (ii): Number of Seats in the 15th Row [1 Mark]

$$a_{15} = 200 + (15-1)(20) = 200 + 14 \times 20 = 200 + 280 = \boxed{\textbf{480 seats}}$$

Sub-part (iii): Total Seating Capacity of All 30 Rows [2 Marks]

$$S_{30} = \frac{30}{2}[2(200) + (30-1)(20)] = 15(400 + 29 \times 20) = 15(400 + 580) = 15 \times 980 = \boxed{\textbf{14,700 seats}}$$

Sub-part (iii) [OR Option]: Total Revenue from First 10 Rows at Rs. 500/Seat [2 Marks]

Total seats in rows 1 to 10:

$$S_{10} = \frac{10}{2}[2(200) + (10-1)(20)] = 5(400 + 180) = 5 \times 580 = 2{,}900 \text{ seats}$$

Revenue = 2,900 × Rs. 500 = **Rs. 14,50,000**.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★